Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/31/2006
PI Name:	Bates, Theodore R. Ph.D.		
Project Title:	Oral Absorption and Drug-drug Interaction Kinetics of Quinolone Antibiotics in an Animal Model of Weightlessness		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77004-4501	**Congressional District:**	18
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Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: NCE to 5/31/07 (info rec'd 6/1/06)		
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	The results of previously reported studies conducted in the tail-suspended (TS) rat model indicated that simulated weightless has little effect on the gastrointestinal absorption and disposition kinetics of the water soluble fluoroquinolone antibiotics, ciprofloxacin and moxifloxacin. However, concomitant oral administration of moxifloxacin with Maalox® (a commercial aluminum hydroxide/magnesium hydroxide-containing antacid preparation) markedly reduced the rate and extent of moxifloxacin absorption in both control and TS rats. Quite interestingly, the oral absorption kinetics of moxifloxacin were insignificantly affected by coadministered Pepto-Bismol® (a commercial bismuth subsalicylate-containing antidiarrheal preparation) in TS rats, but were significantly impaired in control rats. The following important questions arose from these pharmacokinetic studies:		

Task Description:	• What is the effect of administration time interval on the in vivo interaction between moxifloxacin and Maalox® or Pepto-Bismol® and, thus, on the absorption kinetics of moxifloxacin under conditions of weightlessness? • What are the effects of weightlessness on the oral absorption kinetics of a poorly soluble, poorly absorbed antibiotic as compared to the water soluble fluoroquinolone antibiotics? • Are the effects of weightless on the disposition kinetics of drugs affected by gender? We have conducted several experiments to address these questions. Specific Aims of this portion of the Project are: (1) To quantitatively assess whether the time interval between moxifloxacin and an aluminum hydroxide/magnesium hydroxide antacid mixture (Maalox®) or between moxifloxacin and the antidiarrheal agent bismuth subsalicylate (Pepto-Bismol®) oral administrations can affect fluoroquinolone antibiotic absorption kinetics in a tail-suspended microgravity rat model. Our hypothesis is that the administration time interval may alter the effect of weightlessness on the extent of the drug-drug interaction between Maalox® or Pepto-Bismol® and quinolone antibiotics. (2) To study the effect of microgravity on the gastrointestinal (GI) absorption kinetics of the model water-insoluble and inherently poorly absorbed drug griseofulvin using a tail-suspended microgravity rat model. Our hypothesis is that weightlessness may have a significant effect on the rate and extent of absorption (bioavailability) of orally administered griseofulvin. (3) To assess whether simulated weightlessness can affect the disposition kinetics of antipyrine, a marker compound for hepatic oxidative metabolism, and whether gender differences exist. The tail-suspended rat model was used to simulate weightlessness. Our hypothesis is that gender differences in gastrointestinal physiology and/or in the hepatic metabolizing enzyme systems may potentially modify drug absorption and/or disposition during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the planned studies will provide mechanistic information on the currently unknown effects of microgravity on gastrointestinal drug-drug interactions and drug absorption, as well as on the currently unknown effect of gender on drug disposition. Data generated from these animal model kinetic studies should provide useful information pertinent to CPR questions in the area of pharmacology.
Task Progress:	Question 19a: What are the effects of space flight and reduced-G on the absorption, distribution, metabolism, clearance, excretion, clinical efficacy, side effects and drug interactions for medications used in primary, secondary and tertiary prevention of conditions stated in the Space Medicine Condition list? Concomitant oral administration of moxifloxacin and Maalox® to control or tail-suspended rats markedly impaired the rate and extent (bioavailability) of moxifloxacin absorption. This moxifloxacin-Maalox® interaction and resultant impairment of moxifloxacin absorption was statistically significant even when the doses of moxifloxacin and Maalox® was separated by 60 min. However, in tail-suspended rats, the degree of this drug-drug interaction and the effect of the interaction on moxifloxacin absorption was significantly reduced when Maalox® was administered 60 min before moxifloxacin. Oral administration of moxifloxacin with Pepto-Bismol® (dosed at either -5 min or -60 min) to control rats markedly impaired the rate but not the extent of moxifloxacin absorption. This drug-drug interaction involves complexation of moxifloxacin with the trivalent bismuth ion present in Pepto-Bismol® . In striking contrast, oral administration to 3-day tail-suspended rats of moxifloxacin with Pepto-Bismol® at either -5 min or -60 min had no significant effect on either the rate or extent of moxifloxacin absorption. It appears that the condition of weightless affects the equilibrium between free and bismuth ion complexed moxifloxacin. Further mechanistic studies are in progress. Our data also demonstrated that antipyrine clearance was significantly decreased in tail-suspended male rats, which clearly indicates that hepatic oxidative enzymatic function is reduced under simulated weightlessness conditions. These results are consistent with the significant decrease in total hepatic cytochrome P-450 content observed in rats aboard both the Cosmos 1997 spaceflight and the Skylab 3 space station. Quite interesting, simulated weightless had no significant effect on the disposition kinetics of antipyrine in female rats. Question 19d: What effect does space adaptation have on drug bio-availability and how can efficacy be enhanced? The results of our studies on the oral absorption kinetics of griseofulvin, a poorly water soluble drug, provided evidence that simulated weightlessness can appreciably and significantly alter the gastrointestinal absorption characteristics of griseofulvin. Possible reasons for the changes noted in drug bioavailability under weightlessness conditions are related to changes in gastrointestinal motility or gastrointestinal permeability. Further studies on the effect(s) of microgravity on the permeability of the gastrointestinal membrane are warranted.
Bibliography Type:	Description: (Last Updated: 09/04/2007)
Abstracts for Journals and Proceedings	Wei B, Ma J, Liang D, Putcha L, Bates TR. "Pharmacokinetics of Ciprofloxacin in a Simulated Weightlessness Rat Model . " Presented at the Annual Meeting of American Association of Pharmaceutical Scientists (Abstract #T2313), Nashville, TN, November 6-10, 2005. The AAPS Journal, 2005 Nov;7(S2):Abstract #T2313. , Nov-2005
Abstracts for Journals and Proceedings	Liang D, Ma J, Hunter R, Williams B, Putcha L, Bates TR. "Effect of Simulated Weightlessness on the Pharmacokinetics of the Interaction Between Moxifloxacin and Maalox. " Presented at the Annual Meeting of American Association of Pharmaceutical Scientists (Abstract #M1308), Nashville, TN, November 6-10, 2005. The AAPS Journal, 2005 Nov;7(S2):Abstract #M1308. , Nov-2005

Abstracts for Journals and Proceedings	Liang D, Ma J, Wei B, Putcha L, Bates TR. "Effect of Simulated Weightlessness on the Pharmacokinetics of the Interaction Between Moxifloxacin and Pepto-Bismol. " Presented at the Annual Meeting of American Association of Pharmaceutical Scientist (Abstract #M1307), Nashville, TN, November 6-10, 2005. The AAPS Journal, 2005 Nov;7(S2):Abstract #M1307. , Nov-2005